love3.c

```c

/*
#include <stdio.h>

int main()
{
    int num[10] = {12, 4, 5, 1, 9, 13, 11, 19, 54, 34};
    int i, temp;

    // Swap adjacent elements
    for(i = 0; i < 10; i += 2)
    {
        temp = num[i];
        num[i] = num[i + 1];
        num[i + 1] = temp;
    }

    // Print result
    printf("After swapping:\n");
    for(i = 0; i < 10; i++)
    {
        printf("%d ", num[i]);
    }

    return 0;
}
*/



/*
#include <stdio.h>

int main()
{
    int a[5] = {1, 2, 3, 4, 5};
    int b[5];
    int i;

    // Reverse copy
    for(i = 0; i < 5; i++)
    {
        b[i] = a[5 - i - 1];
    }

    // Print original array
    printf("Original array:\n");
    for(i = 0; i < 5; i++)
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    {
        printf("%d ", a[i]);
    }

    // Print reversed array
    printf("\nReversed array:\n");
    for(i = 0; i < 5; i++)
    {
        printf("%d ", b[i]);
    }

    return 0;
}
*/

/*
#include <stdio.h>

int main()
{
    int a[5] = {10, 25, 3, 89, 45};
    int i, max, min;

    max = min = a[0];   // assume first element

    for(i = 1; i < 5; i++)
    {
        if(a[i] > max)
            max = a[i];

        if(a[i] < min)
            min = a[i];
    }

    printf("Maximum = %d\n", max);
    printf("Minimum = %d\n", min);

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[8] = {10, -5, 0, 23, -1, 0, 7, -9};
    int i, pos = 0, neg = 0, zero = 0;

```

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    for(i = 0; i < 8; i++)
    {
        if(a[i] > 0)
            pos++;
        else if(a[i] < 0)
            neg++;
        else
            zero++;
    }

    printf("Positive = %d\n", pos);
    printf("Negative = %d\n", neg);
    printf("Zero = %d\n", zero);

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[5] = {10, 20, 30, 40, 50};
    int i, sum = 0;
    float avg;

    // Calculate sum
    for(i = 0; i < 5; i++)
    {
        sum = sum + a[i];
    }

    // Calculate average
    avg = sum / 5.0;

    printf("Sum = %d\n", sum);
    printf("Average = %f\n", avg);

    return 0;
}
*/



/*
#include <stdio.h>

int main()
```

```c
{
    int a[6] = {10, 25, 30, 45, 50, 60};
    int i, key, found = 0;

    printf("Enter element to search: ");
    scanf("%d", &key);

    for(i = 0; i < 6; i++)
    {
        if(a[i] == key)
        {
            printf("Element found at position %d\n", i + 1);
            found = 1;
            break;
        }
    }

    if(found == 0)
    {
        printf("Element not found\n");
    }

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[5] = {30, 10, 50, 20, 40};
    int i, j, temp;

    for(i = 0; i < 5 - 1; i++)
    {
        for(j = 0; j < 5 - 1 - i; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    printf("Sorted array:\n");
    for(i = 0; i < 5; i++)
```

```c
    {
        printf("%d ", a[i]);
    }

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[5][5] = {
        {11, 1, 7, 9, 7},
        {13, 54, 56, 2, 5},
        {23, 43, 89, 22, 13},
        {14, 15, 17, 16, 19},
        {45, 3, 6, 8, 10}
    };

    int i, j, max;

    max = a[0][0];   // assume first element

    for(i = 0; i < 5; i++)
    {
        for(j = 0; j < 5; j++)
        {
            if(a[i][j] > max)
                max = a[i][j];
        }
    }

    printf("Largest number = %d\n", max);

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int mat1[3][5] = {
        {1, 2, 3, 4, 5},
        {6, 7, 8, 9, 10},
```

```c
        {11, 12, 13, 14, 15}
    };

    int mat2[5][3];
    int i, j;

    // Transpose logic
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 5; j++)
        {
            mat2[j][i] = mat1[i][j];
        }
    }

    // Print transpose
    printf("Transpose matrix:\n");
    for(i = 0; i < 5; i++)
    {
        for(j = 0; j < 3; j++)
        {
            printf("%d\t", mat2[i][j]);
        }
        printf("\n");
    }

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[2][2] = {{1, 2}, {3, 4}};
    int b[2][2] = {{5, 6}, {7, 8}};
    int c[2][2];
    int i, j;

    // Addition
    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            c[i][j] = a[i][j] + b[i][j];
        }
    }

```

```c
    // Print result
    printf("Resultant matrix:\n");
    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[2][2] = {{1, 2}, {3, 4}};
    int b[2][2] = {{5, 6}, {7, 8}};
    int c[2][2] = {0};   // initialize with 0
    int i, j, k;

    // Multiplication
    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            for(k = 0; k < 2; k++)
            {
                c[i][j] += a[i][k] * b[k][j];
            }
        }
    }

    // Print result
    printf("Resultant matrix:\n");
    for(i = 0; i < 2; i++)
    {
        for(j = 0; j < 2; j++)
        {
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }

    return 0;
```

```c
}
*/


/*
#include <stdio.h>

int main()
{
    int a[3][3] = {
        {1, 2, 3},
        {2, 4, 5},
        {3, 5, 6}
    };

    int i, j, flag = 1;

    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            if(a[i][j] != a[j][i])
            {
                flag = 0;
                break;
            }
        }
    }

    if(flag == 1)
        printf("Matrix is symmetric\n");
    else
        printf("Matrix is not symmetric\n");

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[3][3] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

```

```c
    int i, j, sum;

    // Row sums
    printf("Row sums:\n");
    for(i = 0; i < 3; i++)
    {
        sum = 0;
        for(j = 0; j < 3; j++)
        {
            sum += a[i][j];
        }
        printf("Row %d = %d\n", i + 1, sum);
    }

    // Column sums
    printf("\nColumn sums:\n");
    for(j = 0; j < 3; j++)
    {
        sum = 0;
        for(i = 0; i < 3; i++)
        {
            sum += a[i][j];
        }
        printf("Column %d = %d\n", j + 1, sum);
    }

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[3][3] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int i, j;
    int sum1 = 0, sum2 = 0;

    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            if(i == j)
```

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                sum1 += a[i][j];

            if(i + j == 3 - 1)
                sum2 += a[i][j];
        }
    }

    printf("Primary diagonal sum = %d\n", sum1);
    printf("Secondary diagonal sum = %d\n", sum2);

    return 0;
}
*/


/*
#include <stdio.h>

int main()
{
    int a[4][4] = {
        {1, 2, 3, 4},
        {5, 6, 7, 8},
        {9,10,11,12},
        {13,14,15,16}
    };

    int i, j;

    printf("Boundary elements:\n");

    for(i = 0; i < 4; i++)
    {
        for(j = 0; j < 4; j++)
        {
            if(i == 0 || i == 3 || j == 0 || j == 3)
                printf("%d ", a[i][j]);
            else
                printf("  "); // for spacing
        }
        printf("\n");
    }

    return 0;
}
*/


/*
#include <stdio.h>
```

```c

int main()
{
    int a[3][3] = {
        {1, 0, 0},
        {0, 1, 0},
        {0, 0, 1}
    };

    int i, j, flag = 1;

    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            if(i == j && a[i][j] != 1)
                flag = 0;

            if(i != j && a[i][j] != 0)
                flag = 0;
        }
    }

    if(flag == 1)
        printf("Identity matrix\n");
    else
        printf("Not an identity matrix\n");

    return 0;
}
*/


```